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Microscopic structural study on the growth history of granular heaps prepared by the raining method

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Granular heaps are critical in both industrial applications and natural processes, exhibiting complex behaviors that have attracted significant research attention. The stress dip phenomenon observed beneath granular heaps continues to be a topic of significant debate. Current models based on force transmission often assume that the packing is near the isostatic point, overlooking the critical influence of the internal structure and formation history on the mechanical properties of granular heaps. Consequently, these models cannot fully capture for diverse observations. In this study, we experimentally explore the structural evolution of three-dimensional (3D) granular heaps composed of monodisperse spherical particles prepared using the raining method. Our results reveal a continuous transition from a densely packed central core to more loosely packed outer regions, characterized by significant differences in structural properties such as contact number and contact anisotropy. We attribute these structural variations to the differing formation mechanisms during heap growth. Our findings emphasize the substantial influence of the preparation protocols on the internal structure of granular heaps and provide valuable insights into stress distribution within granular materials. This research may contribute to the development of more accurate constitutive relations for granular materials.

I. Introduction

Granular heaps play a vital role in numerous industrial applications and natural phenomena.[1–3] Despite their seemingly simple structure, granular heaps exhibit a wealth of intricate behaviors and complex features that have attracted recurring interest over the years. These include avalanching behavior,[4,5] which is closely related to the well-known concept of self-organized criticality (SOC),[6] the distinctive surface topography of sand dunes in deserts,[7,8] various factors influencing the angle of repose,[9–11] and the stress distribution underneath the pile.[12–14] Specifically, the stress dip observed underneath granular heaps has sparked debate regarding its origin. Edwards and coworkers proposed that there exists asymmetry in the supporting forces exerted on a particle from its supporting particles, leading forces to propagate along chain-like arch structures, which effectively directs stress sideways to reduce the load underneath.[15] Building on this idea, the scalar q model[16] and the more complex oriented stress linearity (OSL) model[12] have been developed. In contrast to conventional elasto-plastic soil mechanics models,[17,18] these models attempt to close the constitutive relations by stress components only without strain. This approach is motivated by the fact that granular particles are rigid and close to the hard-sphere limit,[19] rendering elastic energy negligible when deformations are small. By further assuming that granular systems are nearly isostatic, this new closure scheme leads to a hyperbolic equation for force transmission instead of an elliptic one.[20] This prediction is used to explain the appearance of arches or force chain structures in granular materials and the potential hyperbolic instead of ordinary elliptical force responses, although the experimental verification remains controversial.[21] Concurrently, numerical simulations by Zheng and Yu[22] demonstrated that the appearance of the stress dip depends on the heap preparation history, indicating that its appearance is non-universal and protocol-dependent. This underscores the need for studies that account for both the specific preparation protocol and the history of granular piles to understand their properties. Topić *et al.* performed simulations to investigate the density distribution and angle of repose in three-dimensional (3D)

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heaps of monodisperse spherical particles.[23] Through microscopic structural analysis, their analysis revealed that when the particles are deposited from a small outlet, the resulting heaps exhibit three distinct angles of repose (including an internal angle) and four characteristic density zones, all closely tied to the deposition history. Conversely, a large outlet size results in the merging of the two distinct surface repose angles. These findings further underscore the critical role of the deposition protocol and history in determining the properties of granular heaps. Nevertheless, prior experimental studies on granular heaps have primarily focused on macroscopic properties, such as external angles of repose or base stress distributions, often lacking detailed microscopic information on the internal structure and force distribution. Although photo-elastic experiments offer valuable structural and force information, they are restricted to two dimensions (2D).[21] Therefore, there is a pressing need for systematic experimental investigations of the microscopic structure of 3D granular piles that consider their growth history.

In this paper, we present X-ray tomography experiments to investigate the structural evolution and growth history of 3D heaps composed of monodisperse spherical particles. We investigate the spatial development of packing fraction, number of contacts, and contact anisotropy as a function of heap growth. Our results reveal a continuous transition from a high-density central core to surrounding low-density regions. This spatial gradient arises from the differing deposition mechanisms across the heap, leading to a gradual transformation from a solid-like interior to a loosely packed, liquid–solid boundary state.

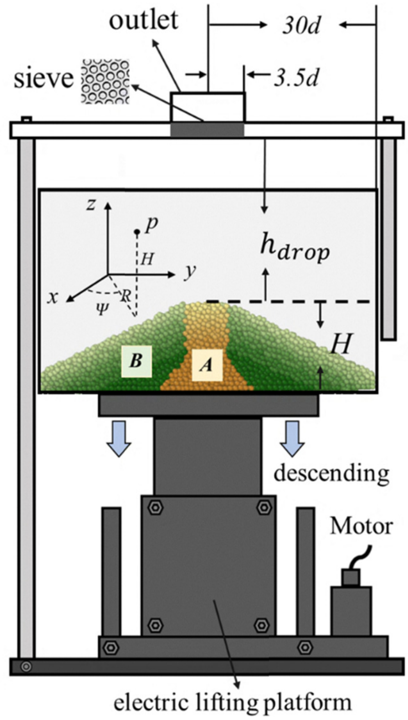

Fig. 1 Schematic of the experimental setup for granular heap formation. The yellow and green shaded regions indicate regions A and B, respectively. The varying shadow intensities reflect particle depths within the heap, with darker shades corresponding to particles located deeper in the pile. An electric lifting platform beneath the container ensures a constant drop height, h_{drop}, during deposition.

II. Experimental methods

We prepare granular heaps by depositing approximately 30 000 monodisperse spherical acrylonitrile butadiene styrene plastic (ABS) beads, each with a diameter of d = 3 mm, into a plexiglass cylindrical container with an inner diameter of 60d. We note that, prior to each experiment, all particles are stored in a grounded metallic container to dissipate any residual surface charge. This is an effective procedure to minimize electrostatic effects in macroscopic granular systems. Moreover, we do not observe abnormal clustering or wall-induced particle aggregation, which are commonly reported in sub-millimeter particulate systems where electrostatic charging is significant,[24] suggesting that electrostatic interactions are negligible under our conditions. Following the standard raining method,[25] the beads are released from a source with an outlet diameter of 7d, positioned directly above the cylindrical container's center. A sieve with a mesh size of 5 mm is placed just below the outlet to ensure a homogeneous "rain" of particles. The drop height h_{drop} is defined as the vertical distance from the sieve to the top of the particle heap. We perform five experimental groups with different h_{drop}: 19d, 27d, 37d, 47d, and 57d. To maintain a constant drop height, h_{drop}, throughout the deposition process, the cylindrical container is mounted on an electric lifting platform with a precision of 10 μm and a stroke of 10 cm. The electric lifting platform descends at a constant rate as particles accumulate, ensuring that the drop height h_{drop} remains fixed. The initial state of the heap is formed by depositing approximately 8600 particles, forming a conical heap with a height of approximately 7d. Subsequently, five additional batches, each containing approximately 4300 particles, are added, bringing the final heap height to approximately 17d. This methodology allows for a systematic investigation of six distinct growth stages of the heap. After each deposition, 3D structures of the packing are obtained through X-ray tomography scans using a medical CT scanner (UEG Medical Group Ltd, 0.2 mm spatial resolution). Using image processing procedures similar to those described in our previous studies,[26,27] we obtain the centroid coordinates of each particle with an error of less than $3 \times 10^{-3}d$. Results for each configuration are averaged over 5 independent realizations to enhance statistical reliability. To mitigate boundary effects, only particles at least 2d away from the heap boundary are included in the analysis. For comparison, tapping experiments are conducted using the same ABS beads (d = 6 mm) in a cylindrical container with an inner diameter of 140 mm. Irregular hemispheres are affixed to both the sidewalls and the bottom of the container to prevent crystallization. The system is subjected to vertical mechanical vibrations of varying intensities, generating a series of granular packings that span the range from random loose packing (RLP) to random close packing (RCP). Further details on the tapping experiments can be found in ref. 28. Each condition is repeated five times, and particles located within 2.5d of the container boundary are excluded from the analysis to further reduce boundary artifacts.

III. Results

A. Two regions within the heap distinguished by packing density

To investigate the structural evolution of the granular heaps during growth, we first examine the azimuthally averaged local volume fraction $\phi = \langle v_p \rangle / \langle v \rangle$, where v_p and v represent the volumes of the particles and their associated Voronoi cells, respectively. Using cylindrical coordinates (R, H, ψ) coaxial with the heap, as shown in Fig. 1, the azimuthal average is calculated over different azimuthal angles for all particles with identical R and H. This averaging is employed due to the conical shape and circular symmetry of the three-dimensional granular heaps in each horizontal plane, with the apex as the center.

Fig. 2(a) shows the spatial distributions of ϕ as a function of radial distance R and height H for six packings, documenting the heap's growth history. The data are coarse-grained by averaging the ϕ of particles within square blocks of size d^2 (see the SI for more details regarding the coarse-graining). In all cases, particles located directly below the outlet, within a cylindrical region of approximately $7d$ diameter, typically have higher ϕ compared to those farther away from the center of the heap, as observed in the simulation of ref. 23. To better investigate the internal evolution of ϕ, we analyze how the ϕ varies with radial distance R at different heights H [inset of Fig. 2(b)]. At $H = 7d$, $10d$ and $13d$, ϕ remains nearly constant for $R < 3.5d$, but drops sharply beyond this radius, reaching a lower plateau at $\phi \approx 0.60$ for $R > 10d$. This behavior allows us to divide the heap into two distinct regions: a central region, A ($R < 3.5d$), and a surrounding region, B ($R > 3.5d$). Interestingly, at a lower height, $H = 4d$, the drop in ϕ occurs at a larger radial distance ($R \approx 6d$), indicating that region A widens near the base, taking the shape of a truncated cone.

Notably, the boundary between these two regions coincides with the outlet radius $3.5d$, suggesting that the observed spatial variation in volume fraction originates from the distinct formation mechanisms in regions A and B. In region A, falling particles directly impact those already deposited, leading to compact, mechanically stable structures. Greater impact energies result in denser arrangements, yielding higher packing fractions. Fig. 2(c) shows that ϕ in region A increases with drop height h_{drop}, supporting this interpretation. This trend is

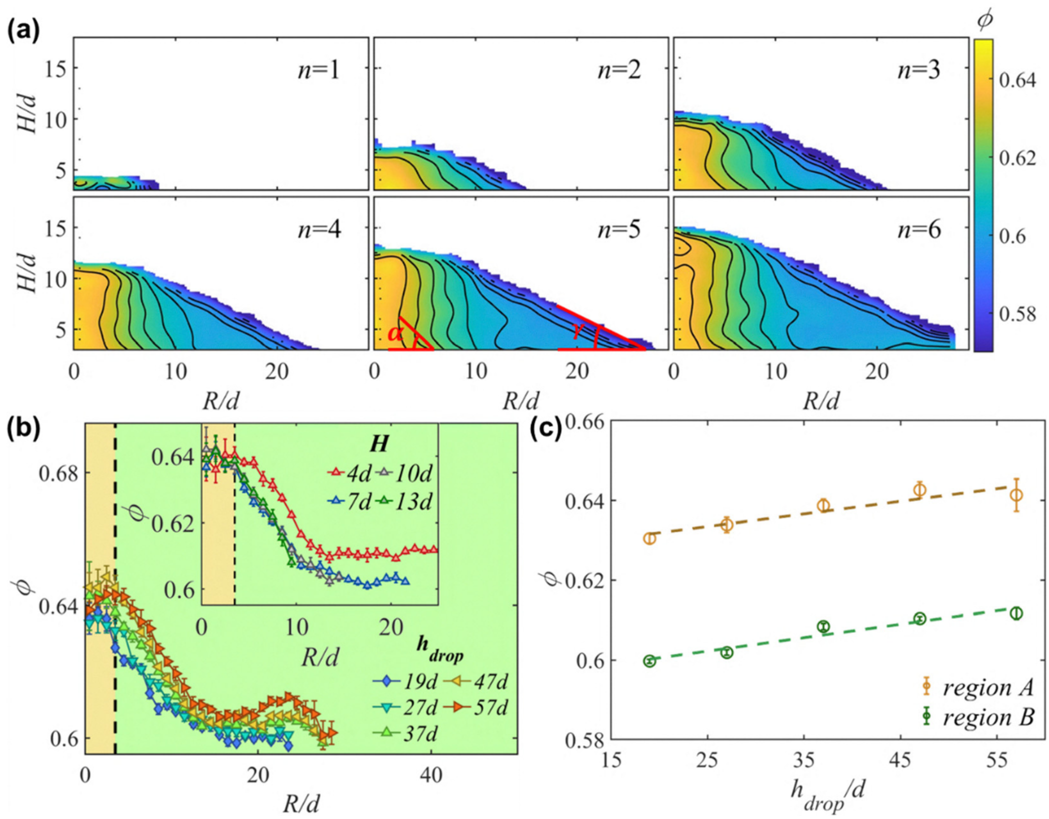

Fig. 2 (a) Spatial distributions of ϕ as a function of radial distance R and height H for different deposition numbers n, shown for drop height h_{drop} = $37d$. (b) Volume fraction ϕ as a function of radial distance R at fixed height $H = 5d$. Different markers represent different drop heights h_{drop}. Inset: Volume fraction ϕ of radial distance R at various heights H for a fixed drop height h_{drop} = $37d$. Background colors distinguish the two structural regions: yellow indicates region A, and green indicates region B. (c) Volume fraction ϕ as a function of drop height h_{drop}. Different colors represent different regions (A and B), with data shown for a fixed drop height h_{drop} = $37d$. Each data point represents the average of five independently repeated experiments, and the error bars represent the standard deviations. The definition of error bars is consistent across all subsequent figures.

consistent with the empirical observations from the pluviation protocol, a standard particle packing preparation protocol in soil mechanics, where particles falling from greater heights typically achieve denser packing.[29–31] By contrast, particles in region B originate from beads that roll down the heap's surface, creating a granular flow along the slope before settling. The final packing structure in this region retains signatures of this preceding flow process. As the beads descend under gravity, they gradually accelerate and eventually form a flow-stabilized layer. In dense granular flows, faster-moving regions generally exhibit lower volume fractions,[32] resulting in more loosely packed structures upon deposition. This phenomenon directly explains the observed drop in ϕ beyond $R = 3.5d$ and the emergence of a plateau for $R > 10d$. These observations highlight the essential role of formation history in shaping the internal structure of granular heaps, an aspect often overlooked in previous studies.[33] We note that our data reveals a more gradual transition between regions A and B, rather than the sharp interface reported in previous studies.[23] This smooth variation likely reflects the progressive onset of granular flow at low velocities, corresponding to a continuous transformation from a stable, solid-like state to a marginally stable state near the liquid–solid boundary, where the distinction between deposition mechanisms becomes subtle.

Additionally, similar to the findings of Topić *et al.*,[23] we observe that the heaps exhibit an external angle of repose $\gamma \approx 25^\circ$ along with a larger internal angle of repose $\alpha \approx 54^\circ$, defined by the boundary between regions A and B in Fig. 2(a). According to Topić's model, particles in region A are packed layer by layer during accumulation, and the internal angle of repose corresponds to the fact that the horizontal top surface requires lateral support. In contrast, the external slope angle of the particle pile θ is determined by the effective macroscopic friction coefficient ($\mu = \tan\theta$) of dense granular flow.

B. Contact number and contact anisotropy

To further elucidate the structural characteristics of the granular heaps, we calculate the average contact number Z and the fabric intensity β of particles in different regions. Ideally, particles in direct contact should have zero surface-to-surface distance. However, due to experimental uncertainties, such as finite X-ray spatial resolution and inaccuracies in determining particle sizes and centroids, particles may be erroneously identified as either overlapping or having gaps. To address this

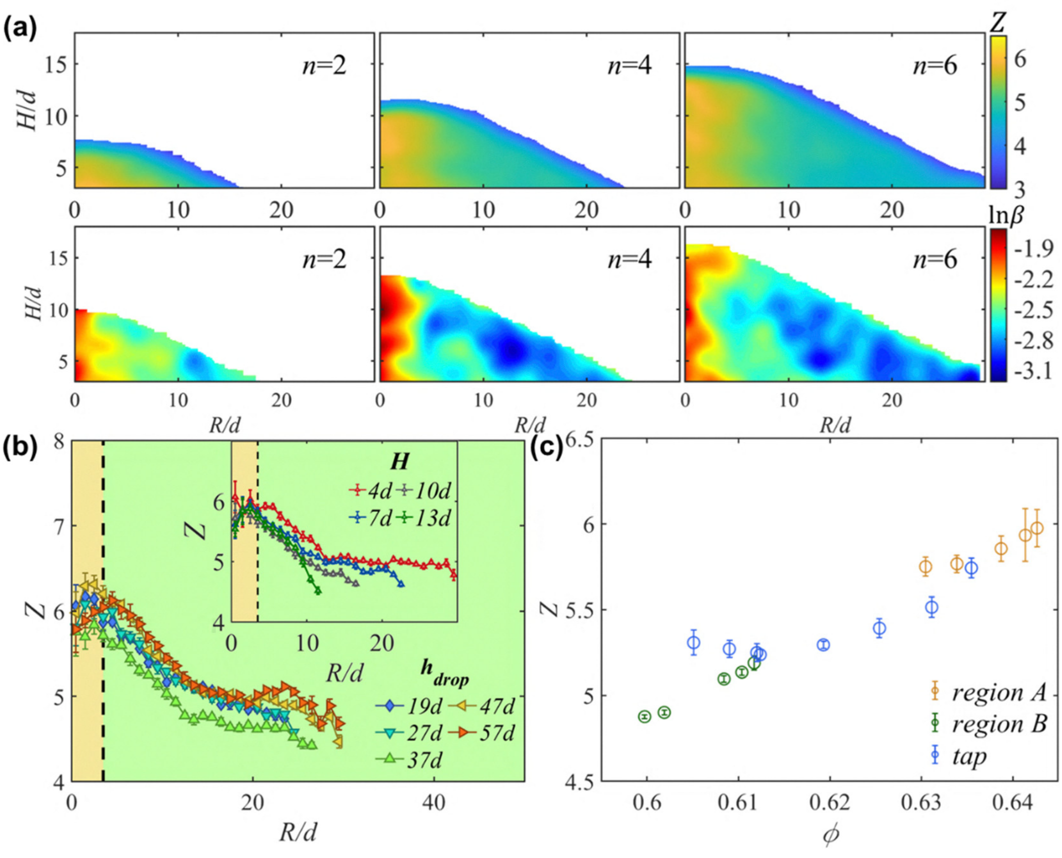

Fig. 3 (a) Spatial distributions of contact number Z and the fabric intensity β as a function of radial distance R and height H for different deposition numbers n = 2, 4, 6, shown for drop height $h_{\mathrm{drop}} = 37d$. (b) Contact number Z as a function of radial distance R at fixed height $H = 5d$. Different markers represent different drop heights h_{drop}. Inset: Contact number Z of radial distance R at various heights H for a fixed drop height $h_{\mathrm{drop}} = 37d$. Background colors distinguish the two structural regions: yellow indicates region A, and green indicates region B. (c) Contact number Z as a function of volume fraction ϕ for a fixed drop height $h_{\mathrm{drop}} = 37d$. Different colors represent data from region A, region B, and the tapping experiments, enabling direct comparison between packings formed in pile experiments and those prepared *via* tapping.

issue, we employ standard procedures for contact determination using a complementary error function fitting approach[34] (see the SI for more details). The contact network is further characterized using the fabric tensor:[35,36]

$$\mathbf{A} = \frac{1}{N_c}\sum_{a=1}^{N_c} \mathbf{n}^a \otimes \mathbf{n}^a - \frac{1}{3}\mathbf{I}, \tag{1}$$

where N_c is the total number of contacts, $\mathbf{n}$ is the unit contact vector between the centers of two contacting particles, and $\mathbf{I}$ is the unit tensor. The contact anisotropy is quantified by the ratio of the maximum and minimum eigenvalues of $\mathbf{A}$, referred to as the fabric intensity β. The distributions of Z (upper) and β (lower) for the growing heap are shown in Fig. 3(a). The contact number Z exhibits a similar evolution trend to the volume fraction ϕ as depicted in Fig. 3(b), with significantly higher values in region A compared to region B. This observation reinforces the hypothesis that the packing structures in region A are more stable than those in region B. Furthermore, the contact number Z is positively correlated with the volume fraction ϕ in both regions of the heap as well as in the tapping experiments [see Fig. 3(c)]. However, regions of the heap with higher ϕ also exhibit larger fabric intensity β, which contrasts with the behavior observed in packings prepared by mechanical tapping [see Fig. 4(a)]. This indicates that the force transmission in the raining-prepared systems is more anisotropic. The resulting directional stress propagation can redirect vertical loads laterally, thereby reducing the central pressure and potentially giving rise to a stress dip, an effect consistent with Edwards' original proposal. Nevertheless, the Voronoi cell volume distributions $P(v_p)$ of the packings, prepared using different methods but exhibiting similar ϕ, are nearly identical [inset of Fig. 4(a)]. This indicates that, despite variations in local contact structures, these systems share the same "compactivity" or "effective temperature" within the Edwards volume ensemble.[36] This further implies that the physics at the contact and particle scales are weakly coupled in these systems, distinguishing them from those near isostaticity.

C. History-dependent spatial distribution of contact points

To investigate the disparity in contact distributions, we compute the spatial distribution of contact points on the particle surface, denoted as $f_i(\mathbf{r}) = \delta(\mathbf{r} - \mathbf{r}_j)$, where $\mathbf{r}_j(r, \theta, \varphi)$ represents the contact point position in spherical coordinates defined for bead i. For averaging purposes, we define $\varphi = 0$ as the direction from the particle center to the heap's center in spherical coordinates. The function $f_i(\mathbf{r})$ is normalized by $\iint_{S_i} f_i(\mathbf{r})\mathrm{d}s = Z_i$, where S_i is the surface area of bead i in units of d^2, and Z_i is its contact number. $\langle f(\mathbf{r})\rangle$ represents the average of $f_i(z)$ across all particles in region A or B [Fig. 4(c)–(h)]. The front views of $\langle f(\mathbf{r})\rangle$ in Fig. 4(c)–(h) are taken at $\varphi = 90°$, with the left-hand side oriented towards the heap center. The probability distribution functions (PDFs) of contact points $p(\theta)$ for particles in regions A and B are shown in Fig. 4(b). For comparison, we also calculate $p(\theta)$ for

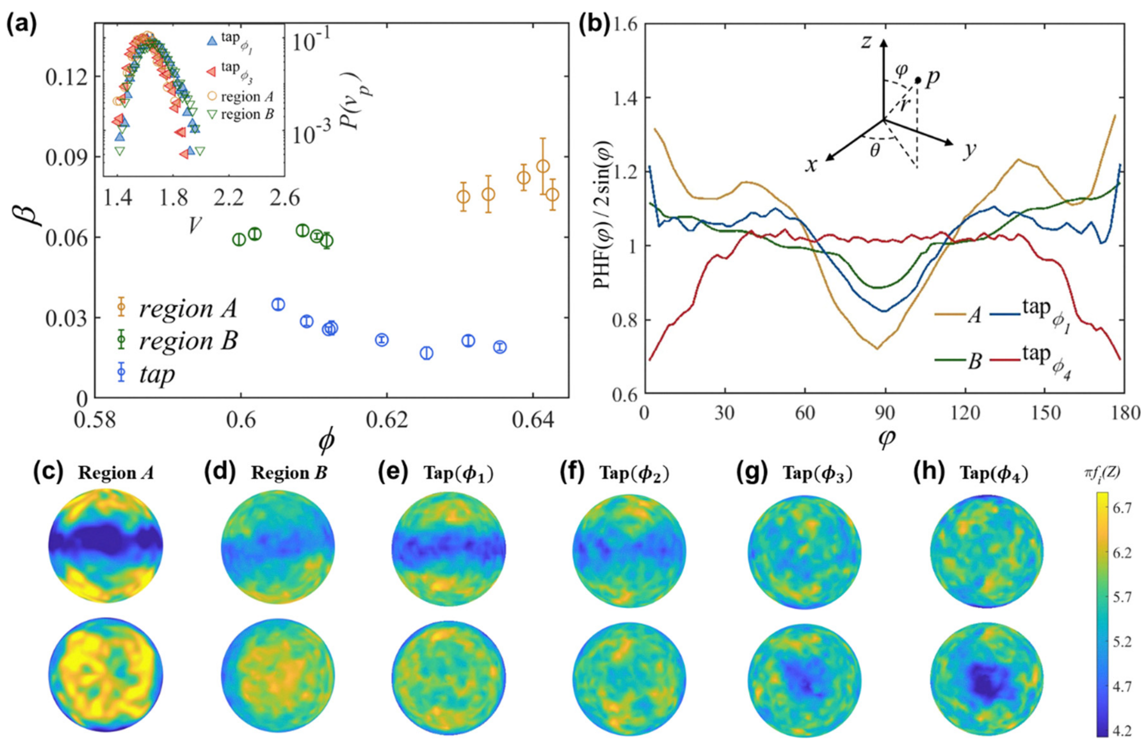

Fig. 4 (a) Fabric intensity β as a function of volume fraction ϕ in region A, region B and tapped packings. Inset: $P(v_p)$ of Voronoi cell volume in regions A and B and tapped packings with $\phi_1 = 0.605$ and $\phi_3 = 0.626$. (b) PDFs of contact points (PDF(φ)) as a function of φ for particles in regions A and B and tapped packings with $\phi_1 = 0.605$ and $\phi_4 = 0.636$. (c)–(h) The spatial distributions of $f_i(z)$ on the particle surface in regions A and B and tapped packings with $\phi_1 = 0.605$, $\phi_2 = 0.612$, $\phi_3 = 0.626$ and $\phi_4 = 0.631$. The first line (second line) displays the sectional view (bottom view). The spatial distribution $f_i(z)$ of contact points is normalized by the spherical surface area.

packings obtained from vertical tapping experiments with $\phi_1 = 0.605$ and $\phi_4 = 0.631$ [blue and red curves in Fig. 4(b)] and analyze $\langle f(\mathbf{r})\rangle$ for packings with $\phi_1 = 0.605$, $\phi_2 = 0.612$, $\phi_3 = 0.626$, and $\phi_4 = 0.631$ [Fig. 4(e)–(h)]. In regions A and B, we observe that contact points tend to concentrate at the top and bottom of the particles, with fewer contacts distributed near the equator. A similar pattern emerges in tapping experiments with high tap intensity Γ, *i.e.*, low ϕ. This phenomenon is likely due to vertical contacts offering greater mechanical stability under vertical impact compared to horizontal contacts. However, in tapped packings, as Γ decreases, the contacts become more uniformly distributed across the particle surface [Fig. 4(e)–(h)]. This occurs because, at lower Γ, particles have lower kinetic energy, which promotes the formation of stable configurations governed by lateral frictional forces, thereby leading to a more uniform distribution of contacts. Note that in region A of the heap, despite with ϕ similar to that of Fig. 4(g), the contact distributions are significantly different. This is most likely due to the preparation method. Unlike the tapped packings, particles in the raining method are deposited sequentially. As a result, collective structures like bridges, which rely on mutual support, are less likely to form during the process. Instead, to withstand the vertical impacts, contacts are more likely to be distributed near the poles. Overall, this suggests that the contact structure is highly dependent on the preparation protocol and does not exhibit a one-to-one correspondence with ϕ. Additionally, in region B, the "blue band", which denotes fewer contacts near the particle equator, appears slightly tilted [Fig. 4(d)]. This occurs because, during heap formation, the existing structures are formed to withstand particles flowing down the slope, leading them to form contacts and stabilized structures that are tilted to accommodate this specific impact geometry. This further emphasizes the strong connection between the internal contact structures of the heap and its growth history.

IV. Conclusion

In summary, we use X-ray tomography to investigate the internal structure and growth history of 3D granular heaps composed of monodisperse spherical particles prepared *via* the raining method. Our analysis of the spatial distribution of the packing fraction reveals that these heaps can be characterized by two distinct regions, A and B, with clear differences in structural properties such as the contact number and contact anisotropy. These regions emerge due to different formation mechanisms during heap growth: while region A forms through direct impact and compaction beneath the outlet, region B develops from particles involved in surface flow prior to deposition. The contrasting internal structures of heaps prepared by the raining and tapping methods underscore the strong dependence of granular packing configurations on the preparation protocol.

More broadly, the history dependence of packing configurations revealed in this study is a general feature of disordered materials, including colloidal glasses, foams, and emulsions. In such systems, the final structure is not solely determined by an external control parameter but is also strongly influenced by the kinetic pathway during preparation. These history-dependent structural variations generate internal heterogeneity, which in turn governs stress propagation, flow onset, and mechanical response. Our results clearly demonstrate how deposition dynamics imprint structural anisotropy into the material, offering a useful reference for understanding soft matter phenomena such as irreversibility and aging. Additionally, we note that the present experiments are conducted using millimeter-scale plastic particles, under conditions where thermal agitation and long-range electrostatic interactions are negligible. In systems composed of smaller particles, such as colloidal suspensions or dielectric powders, additional effects like Brownian motion and substrate-induced electrostatic charging may influence the formation of structure. For example, charge accumulation at container walls may lead to particle aggregation or long-range correlations,[37] which are not observed in our system. These interactions may further affect the sensitivity of packing structures to preparation history. Despite these distinctions, we believe the physical insights provided in this work may serve as a valuable reference for analyzing structure–history relationships in a wide range of soft matter systems.

Returning to the controversy surrounding the stress dip underneath the heap, our study clearly demonstrates that preparation history is pivotal in determining the final force structure of the system. Consequently, the oversimplified assumption that such systems are isostatic fails to account for such diverse observations. Moreover, it is experimentally observed that granular packings typically do not reside at the isostatic point.[28] Nevertheless, in our opinion, the debate between force transmission models and the traditional elasto-plastic soil mechanics models remains far from resolved. Notably, even when systems are far from isostatic, collective structures such as bridges can still form. This supports Edwards' original proposal that constitutive relations could be closed using stress components alone, as his explanation for the occurrence of force chains does not require isostaticity. Additionally, it remains valid that elastic energy plays a negligible role in granular materials at low pressure, highlighting the need for developing constitutive models that are independent of strain. Indeed, within Edwards' volume ensemble, rigidity arises not from elastic restoring forces but from the underlying packing configurations.[38] In this framework, mechanically stable configurations, which are generally thought to be coupled to the stress transmission pathways in granular packings, can, in principle, be described statistically. This allows for a constitutive relation that does not rely on strain and does not require isostaticity, potentially offering a resolution to this long-standing issue.

Conflicts of interest

There are no conflicts to declare.

Data availability

The data supporting this article have been included as part of the SI. The supplementary information contains additional figures, and raw experimental data. See DOI: https://doi.org/10.1039/d5sm00506j.

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